

UNIVERSITY NAME

*Probability and Statistics — Final Exam**Year: 2024***Exercise 1:**

In a factory, two machines M_1 and M_2 are used together to produce parts. Over a given period, the probabilities of breaking down are 0.01 and 0.008, respectively. Moreover, the probability of the event “Machine M_2 is broken down given that M_1 is broken down” is equal to 0.4.

1. What is the probability that both machines break down at the same time?
2. What is the probability that at least one machine is working?

Answer Area

Let A = "Machine M_1 breaks down", B = "Machine M_2 breaks down". Given: $P(A) = 0.01$, $P(B) = 0.008$, $P(B | A) = 0.4$.

1. Probability both break down:

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.4 \times 0.01 = 0.004.$$

2. Probability that at least one machine is working:

This is the complement of both breaking down:

$$1 - P(A \cap B) = 1 - 0.004 = 0.996.$$

Exercise 2:

Let (X, Y) be a random pair with probability density function

$$f(x, y) = \begin{cases} axy^2 & \text{if } 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

1. Calculate a .
2. Calculate the marginal densities of X and Y .
3. Give the cumulative distribution function of Y .
4. Compute the expected value of Y .
5. Calculate $P(X > \frac{1}{2}Y)$.

Answer Area

1. Calculate
- a
- :

The joint density must integrate to 1 over the region $0 \leq x \leq y \leq 1$:

$$\int_{y=0}^1 \int_{x=0}^y axy^2 dx dy = 1.$$

Inner integral:

$$\int_0^y axy^2 dx = ay^2 \left[\frac{x^2}{2} \right]_0^y = ay^2 \cdot \frac{y^2}{2} = \frac{a}{2}y^4.$$

Then:

$$\int_0^1 \frac{a}{2}y^4 dy = \frac{a}{2} \cdot \left[\frac{y^5}{5} \right]_0^1 = \frac{a}{2} \cdot \frac{1}{5} = \frac{a}{10}.$$

Set $\frac{a}{10} = 1 \Rightarrow a = 10$.

2. Marginal densities:

For $0 \leq y \leq 1$:

$$f_Y(y) = \int_{x=0}^y 10xy^2 dx = 10y^2 \cdot \frac{y^2}{2} = 5y^4.$$

For $0 \leq x \leq 1$:

$$f_X(x) = \int_{y=x}^1 10xy^2 dy = 10x \left[\frac{y^3}{3} \right]_{y=x}^1 = 10x \left(\frac{1}{3} - \frac{x^3}{3} \right) = \frac{10}{3}x(1 - x^3).$$

3. CDF of
- Y
- :

For $0 \leq t \leq 1$:

$$F_Y(t) = \int_0^t 5y^4 dy = [y^5]_0^t = t^5.$$

$$F_Y(t) = \begin{cases} 0 & t < 0, \\ t^5 & 0 \leq t \leq 1, \\ 1 & t > 1. \end{cases}$$

4. Expected value of
- Y
- :

$$E(Y) = \int_0^1 y \cdot 5y^4 dy = 5 \int_0^1 y^5 dy = 5 \cdot \frac{1}{6} = \frac{5}{6}.$$

5. Calculate
- $P(X > \frac{1}{2}Y)$
- :

Region condition: $0 \leq x \leq y \leq 1$ and $x > y/2$ (i.e., $y < 2x$). Since $x \leq y$, the inequality $x > y/2$ is equivalent to $y < 2x$ and $x \leq y$. Thus $x \leq y < 2x$. Also $y \leq 1$.

Integrate over x from 0 to 1, and for each x , y runs from x to $\min(2x, 1)$.

For $0 \leq x \leq 1/2$, $2x \leq 1$, so $y \in [x, 2x]$.

For $1/2 < x \leq 1$, $2x > 1$, so $y \in [x, 1]$.

$$P = \int_{x=0}^{1/2} \int_{y=x}^{2x} 10xy^2 dy dx + \int_{x=1/2}^1 \int_{y=x}^1 10xy^2 dy dx.$$

First inner integral:

$$\int_x^{2x} 10xy^2 dy = 10x \left[\frac{y^3}{3} \right]_x^{2x} = 10x \left(\frac{8x^3}{3} - \frac{x^3}{3} \right) = 10x \cdot \frac{7x^3}{3} = \frac{70}{3}x^4.$$

First outer:

$$\int_0^{1/2} \frac{70}{3}x^4 dx = \frac{70}{3} \cdot \frac{(1/2)^5}{5} = \frac{70}{3} \cdot \frac{1}{160} = \frac{70}{480} = \frac{7}{48}.$$

Second inner integral:

$$\int_x^1 10xy^2 dy = 10x \left(\frac{1}{3} - \frac{x^3}{3} \right) = \frac{10}{3}x(1 - x^3).$$

Second outer:

$$\int_{1/2}^1 \frac{10}{3}x(1 - x^3) dx = \frac{10}{3} \int_{1/2}^1 (x - x^4) dx = \frac{10}{3} \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_{1/2}^1.$$

Evaluate at 1: $\frac{1}{2} - \frac{1}{5} = \frac{3}{10}$.

At $x = 1/2$: $\frac{(1/2)^2}{2} - \frac{(1/2)^5}{5} = \frac{1/4}{2} - \frac{1/32}{5} = \frac{1}{8} - \frac{1}{160} = \frac{20}{160} - \frac{1}{160} = \frac{19}{160}$.

Difference: $\frac{3}{10} - \frac{19}{160} = \frac{48}{160} - \frac{19}{160} = \frac{29}{160}$.

Multiply by $\frac{10}{3}$: $\frac{10}{3} \cdot \frac{29}{160} = \frac{290}{480} = \frac{29}{48}$.

Sum: $\frac{7}{48} + \frac{29}{48} = \frac{36}{48} = \frac{3}{4}$.

Exercise 3:

A company wants to hire an executive. n candidates apply for the position. Each of them takes a test in turn, and the first one who passes the test is hired. Let p be the probability that a candidate passes the test. Let X be the random variable defined by:

$$X = \begin{cases} k & \text{if the } k^{\text{th}} \text{ candidate is hired,} \\ n+1 & \text{if no candidate is hired,} \end{cases}$$

1. Give the probability distribution of X .
2. Calculate $E(X)$.
3. Find the minimum value of p such that there is more than a 50% chance of hiring one of the candidates.

Answer Area

1. Probability distribution of X :

For $k = 1, 2, \dots, n$: First $k - 1$ candidates fail, k -th passes:

$$P(X = k) = (1 - p)^{k-1}p.$$

If no one passes: all n fail:

$$P(X = n + 1) = (1 - p)^n.$$

2. Calculate $E(X)$:

$$E(X) = \sum_{k=1}^n k(1 - p)^{k-1}p + (n + 1)(1 - p)^n.$$

Let $q = 1 - p$. The sum $\sum_{k=1}^n kq^{k-1}p = p \cdot \frac{1 - (n+1)q^n + nq^{n+1}}{(1-q)^2}$ (using formula for finite geometric series derivative). Since $1 - q = p$:

$$p \cdot \frac{1 - (n+1)q^n + nq^{n+1}}{p^2} = \frac{1 - (n+1)q^n + nq^{n+1}}{p}.$$

Then:

$$E(X) = \frac{1 - (n+1)q^n + nq^{n+1}}{p} + (n+1)q^n.$$

Simplify:

$$E(X) = \frac{1}{p} - \frac{(n+1)q^n}{p} + \frac{nq^{n+1}}{p} + (n+1)q^n.$$

Combine terms with $(n+1)q^n$:

$$(n+1)q^n - \frac{(n+1)q^n}{p} = (n+1)q^n \left(1 - \frac{1}{p}\right) = (n+1)q^n \cdot \frac{p-1}{p} = -(n+1)q^n \cdot \frac{q}{p}.$$

So:

$$E(X) = \frac{1}{p} + \frac{nq^{n+1}}{p} - (n+1)q^n \cdot \frac{q}{p} = \frac{1}{p} + \frac{q^{n+1}}{p} [n - (n+1)] = \frac{1}{p} - \frac{q^{n+1}}{p}.$$

Thus:

$$E(X) = \frac{1 - (1 - p)^{n+1}}{p}.$$

3. Minimum p such that $P(\text{hire}) > 0.5$:

Probability of hiring = $1 - P(X = n + 1) = 1 - (1 - p)^n$. Require:

$$1 - (1 - p)^n > 0.5 \quad \Rightarrow \quad (1 - p)^n < 0.5.$$

Take natural log (since $1 - p > 0$):

$$n \ln(1 - p) < \ln(0.5) \quad \Rightarrow \quad \ln(1 - p) < \frac{\ln(0.5)}{n}.$$

Since $\ln(1-p) < 0$, we have:

$$1-p > e^{\frac{\ln(0.5)}{n}} = 0.5^{1/n}.$$

Thus:

$$p < 1 - 0.5^{1/n} \quad (\text{but wait, this flips inequality? Let's check carefully}).$$

Actually: $(1-p)^n < 0.5 \Rightarrow 1-p < 0.5^{1/n}$ (since n th root is increasing). So:

$$p > 1 - 0.5^{1/n}.$$